

Check Out Tools from the Crib SOP (1x Episode: five tools, in situ)

One episode checks out **five tools** from the tool crib. It works from one pick list. For each tool it does three things, always in the same order: pull the tool from its shelf, log it on the clipboard, and lay it in its cut-out on the cart. The episode ends when all five tools lie in the shadow foam, all five lines on the list are ticked, and the pen and the clipboard are back where they started. Never split one list into two episodes.

The base is **passive**. It has no motor. You push it by hand up to the crib shelf and lock it there. The cart is pushed up beside the base by hand, and its wheels are locked before recording starts. Nothing is carried away from the crib. Everything the episode touches is already there when recording starts: the shelf with the tools on it, the clipboard with the list and the pen on the cart ledge, and the foam tray on top of the cart.

The crib is used **as found**. Each list tool lies in its own slot on the shelf, with its name tag at the front of the slot. The clipboard lies flat on the cart ledge with the list facing up. The pen lies in its groove next to the clipboard. All five cut-outs in the foam are empty. The episode ends with all five tools seated in the foam, all five boxes ticked, the pen in its groove, and the clipboard flat where it started.

This is an in situ task. Four rules come from that. First, **the shelf and the cart never move**. You never lean on, push, pull, or lift a shelf, a shelf edge, the cart, or the foam tray. Second, **one tool at a time**. A tool leaves its slot only after the tool before it is seated in the foam. Third, **every tool is logged before it is laid down**. The tick comes before the foam, every time. Fourth, **a tool goes only in its own cut-out**. Never in another tool's cut-out. Never on top of the foam. Never on the ledge. Never on the floor.

Only list tools are touched. Other tools on the same shelf are not on the list. Nobody lifts, pushes, or moves them. If a tool on the list is not in its slot, you do not look for it somewhere else. You stop and report it.

The order never changes: **read the next line, pull that tool, tick its box, lay it in its cut-out**. Line 1 first, then line 2, then line 3, then line 4, then line 5.

The grippers never swap jobs in the middle of a tool. **One gripper carries the tool**. It holds the tool from the shelf all the way to the foam. **The other gripper works the pen** and nothing else. It never touches a tool. Nothing is passed from one gripper to the other.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

- **cell_type:** bimanual

Base positioning

The base is passive. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over. The cart is parked the same way.

1. Push the base by hand up to the crib shelf. Stop it square to the front edge of the shelf, so the shelf edge runs straight across the frame and both ends look the same distance away.
2. Stop it in the middle of the shelf bay that holds the five list tools, so the slot at one end and the slot at the other end are the same distance from the middle of the base.
3. Stop it close enough that a gripper reaches the back of every list slot without stretching the arm. Stop it far enough that no arm, wrist, or part of the base touches the shelf edge or the shelf above while both arms work.
4. Push the cart by hand up beside the base. Stop it square to the base, with the foam tray flat and the ledge facing the shelf. Lock both cart wheels.
5. Check the reach on the cart. A gripper reaches every one of the five cut-outs, the tick boxes on the clipboard, and the pen in its groove. It does all of this without stretching and without hitting the shelf or the other arm.
6. Check the height. With the base parked, a gripper lifts the tool in the highest list slot just off the shelf and pulls it straight forward. The wrist and the forearm must not hit the shelf above.
7. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn. Push the cart once, hard, the same way. It must not roll either.
8. If any line from 1 to 7 fails, push the base or the cart to a new spot by hand and start again at line 1. Do not work at a shelf the arms cannot reach easily.

The base and the cart stay locked and still for the whole episode. Nothing moves them. No arm leans on the shelf or the cart hard enough to push them. Nobody touches them by hand. A base or a cart that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera points at the middle of the shelf bay. The frame shows all five list slots with their name tags, the clipboard, the pen in its groove, and the whole foam tray with all five cut-outs.
3. The camera sees the shelf slots from the front, so you can see which slot still holds a tool and which slot is empty.

4. The camera sees the list on the clipboard, so you can read which boxes are ticked and which are empty.
5. The camera sees the foam tray from above and in front, so you can see each cut-out filled or empty, and see if a tool sits flat in it.
6. Both arms are at home with the grippers open.
7. A gripper reaches every list slot on the shelf, the pen in its groove, every tick box on the list, and every cut-out in the foam. It reaches all of them without stretching to its limit.
8. A gripper lifts a tool just off the shelf and carries it low and level to the cart. It never lifts a tool up to the shelf above. It never carries one over the clipboard or over another slot.
9. The two arms do not hit each other. The arm at the pen stays out of the path of the arm that carries the tool.
10. If the arms cannot reach a place, park the base and the cart again with the Base positioning steps until lines 7 to 9 are all true.

Materials checklist

1. The crib shelf is one bay of open shelving, fixed to the floor. Nobody moves it, leans on it, or stands anything on its edge.
2. The five list tools for this episode are on the shelf: a **claw hammer**, an **adjustable wrench**, a pair of **pliers**, a **flat screwdriver**, and a **tape measure**. Each one has a handle to hold. The tape measure is held by its body.
3. Each list tool lies flat in its own slot on the shelf, with its handle toward the front. A slot is a painted outline on the shelf the shape of the tool. There is open shelf between one slot and the next.
4. Each slot has a name tag on the shelf edge at the front of the slot. The name on the tag reads the same as the name on the list line for that tool.
5. Other tools may lie in other slots on the same shelf. They are not on the list. Their names are not on the list. Nobody touches them.
6. No list slot is empty at the start. No list tool lies in the wrong slot. If it does, fix it before recording.
7. The pick list is one sheet of paper held on a clipboard. It has five lines, one under the other. Each line has one tick box at the left and one tool name to the right of the box. All five boxes are empty at the start.
8. The five names on the list are in the order to pull them. Line 1 is at the top. Line 5 is at the bottom.
9. The clipboard lies flat on the cart ledge, list facing up, with the top of the list toward the shelf. Nobody lifts the clipboard or turns it during the episode.
10. The pen is one felt pen with no cap. It makes a dark mark with a light touch. It lies flat in its groove on the ledge, next to the clipboard, with its tip toward the shelf.

11. The cart is a flat cart on four wheels. Its top is the foam tray. Its front, facing the shelf, is a flat ledge for the clipboard and the pen groove. Both wheel locks are on.
12. The shadow foam is one foam block that fills the tray. It has five cut-outs, one for each list tool. Each cut-out is the shape of its tool. The foam is dark on top and bright at the bottom of each cut-out, so an empty cut-out shows bright.
13. Each cut-out has its tool name printed on the foam beside it. The name reads the same as the tag and the list line.
14. All five cut-outs are empty at the start. Nothing lies on top of the foam. Nothing lies on the ledge except the clipboard and the pen.
15. The floor around the base and the cart is clear and dry.

Workspace layout

Nothing is marked or taped out for the robot. You judge every place below by eye, from the shelf and the cart themselves.

- **Shelf bay:** the shelf the base is parked at. It holds the five list slots. It never moves.
- **List slot:** one painted outline on the shelf with a name tag at its front. Each list tool starts in one. It is empty at the end.
- **Other slots:** slots on the same shelf that hold tools not on the list. Never touched.
- **Cart:** parked beside the base, wheels locked. Never pushed.
- **Ledge:** the flat front of the cart. It holds the clipboard and the pen groove. Nothing else is ever put down on it.
- **Clipboard:** lies flat on the ledge, list facing up. Never lifted.
- **Pen groove:** the shallow groove on the ledge next to the clipboard. The pen starts and ends in it.
- **Foam tray:** the top of the cart. It holds the shadow foam with the five cut-outs.
- **Cut-out:** one tool-shaped hole in the foam, with the tool name beside it. Each list tool ends in its own one.
- **Carry path:** straight forward out of the slot, then low and level across to the cart. Never over another slot. Never over the clipboard. Never up toward the shelf above.

Gripper rules

- **The tool gripper:** the gripper that closes on a tool. It lifts the tool out of its slot, carries it to the cart, holds it still while its line is ticked, and lays it in its cut-out. For one tool it is always the same gripper. It never touches the pen.
- **The pen gripper:** the gripper that closes on the pen. It lifts the pen, makes one tick, and lays the pen back in its groove. It never touches a tool. It never touches the clipboard.
- Nothing is passed from one gripper to the other. Only one tool is off the shelf at a time.

Vocabulary

- **List:** the sheet of paper on the clipboard. It has five lines.
- **Line:** one row on the list. One tick box at the left and one tool name to its right.
- **Next line:** the top line on the list whose box is still empty. The next tool is always the tool named on the next line.
- **Name tag:** the small tag on the shelf edge at the front of a slot. It names the tool that belongs in that slot.
- **Matching:** the name on the list line, the name on the tag, and the name beside the cut-out all read the same. The tool in the gripper is that tool.
- **Handle:** the part of the tool you hold. The hammer handle. The wrench handle. Both handles of the pliers together. The screwdriver handle. The body of the tape measure. Never the head, the jaws, the blade, or the tip.
- **Just off the shelf:** the tool has left the shelf, and no higher. Its underside is clear of the paint, and it is not lifted toward the shelf above.
- **Low carry:** you lift the tool just off the shelf, pull it straight forward out of its slot, and carry it low and level across to the cart. You never lift it toward the shelf above. You never carry it over another slot or over the clipboard.
- **Held still:** the tool hangs level and does not swing, turn, or move while the tick is made.
- **Tick:** one short mark inside the tick box. It is inside the box and touches nothing outside the box. One mark, not two. Not a scribble.
- **Ticked line:** a line with a tick in its box. Its tool has been pulled.
- **Seated:** the tool lies flat, all the way down in its own cut-out, with its handle where the handle shape is. No part of it sticks up above the top of the foam. It does not move when the gripper opens.
- **Wrong cut-out:** any cut-out other than the one with this tool's name beside it.
- **Back and clear:** the arm is pulled back, so no part of it is over the shelf, the ledge, or the foam tray.

Steps

Step 1: Find the next tool

Goal: you know which tool comes next, and you know which slot it lies in.

1.1 Read the next line

- Look at the list on the clipboard. Find the **next line:** the top line with an empty box.
- Read the tool name on that line.
- Do not touch the clipboard. Do not touch the pen yet.

1.2 Find its slot

- Look at the name tags along the shelf edge. Find the tag with the same name.
- Check that a tool lies in the slot behind that tag, and that it is the tool named.

Check: the name on the next line and the name on the tag read the same. The tool in that slot is that tool. If there is no tag with that name, or the slot is empty, stop and report it. Do not take a tool from another slot instead.

Expected state: both grippers are empty and back and clear. You know the next tool and its slot. Nothing on the shelf or the cart has been touched.

Step 2: Pull the tool from its slot

Goal: the tool hangs level in the tool gripper, held still over the cart, beside the clipboard.

2.1 Close on the handle

- Bring the tool gripper in from the front, low over the shelf.
- Close on the **handle** in the middle. Close all the way before you lift.
- Do not close on the head, the jaws, the blade, or the tip.

2.2 Lift it out

- Lift the tool **just off the shelf**. Its underside leaves the paint, and no higher.
- Pull it straight forward out of the slot. Do not swing it. Do not drag it along the shelf.
- Do not touch the tool in the slot next to it. Do not touch the name tag.

Check: the tool hangs level in the tool gripper. Its slot is empty and its tag is still in place. The tools in the slots next to it have not moved. No part of the arm is near the shelf above. If the tool leans or swings, lay it back down in its slot, take the handle again in the middle, and lift again.

2.3 Carry it to the cart

- Carry the tool across to the cart in a **low carry**. Keep it level the whole way.
- Do not carry it over another slot. Do not carry it over the clipboard.
- Stop over the foam tray, beside the clipboard, with the tool held still just above the foam. Do not put it down.

Expected state: the tool hangs level and held still over the cart. Its slot on the shelf is empty. Its line on the list is not ticked yet. The pen is still in its groove.

Step 3: Log the tool on the clipboard

Goal: the tool's line has one tick in its box, and the pen is back in its groove.

3.1 Match the tool to the line

- Look at the tool in the tool gripper. Look at the name on the next line.
- They must be the same tool. The tick goes on the next line and no other line.

Check: the tool in the gripper is the tool named on the next line. If it is not, do not tick anything. Carry the tool back to its own slot in a low carry, lay it flat inside its outline, open the gripper, and start again at Step 1.

3.2 Pick up the pen

- With the pen gripper, close on the pen in the middle.
- Lift the pen just off the groove.
- Do not touch the clipboard. Do not touch the tool.

3.3 Tick the box

- Bring the pen tip down inside the box on the next line.
- Make **one short mark** inside the box. Press lightly.
- Lift the pen straight up off the paper.

Check: there is one dark mark inside the box on the right line. Every box above it already has a tick. Every box below it is still empty. Nothing is marked outside the box. If the mark missed the box, or it is too faint to see from the front, make one more short mark inside the box. Do not scribble. Do not cross anything out. Say so when you hand the station back.

3.4 Put the pen back

- Lay the pen flat in its groove, tip toward the shelf.
- Open the pen gripper and pull it back and clear.

Expected state: the tool's line is ticked. The pen lies in its groove. The tool gripper still holds the tool still over the foam tray. The clipboard has not moved.

Step 4: Lay the tool in its cut-out

Goal: the tool is seated in its own cut-out, and both grippers are back and clear.

4.1 Find its cut-out

- Read the names printed beside the cut-outs. Find the one with this tool's name.
- Look at the shape of that cut-out. It is the shape of the tool in the gripper.

Check: the name beside the cut-out reads the same as the tag and the list line. The shape matches the tool. If no cut-out matches, do not push the tool into another one. Hold the tool still and report it.

4.2 Line it up

- Move the tool low over its cut-out.
- Turn it so its shape lies over the shape of the hole, with the handle over the handle shape.
- Keep it level. Keep it just above the foam.

4.3 Lower it in

- Lower the tool straight down into the cut-out until it rests on the bottom.
- Do not push it down. Do not force it. If it will not go in, lift it, turn it a little, and lower it again.
- Open the tool gripper and pull it back and clear.

Check: the tool is **seated**. It lies flat, all the way down, handle where the handle shape is, with no part above the top of the foam, and it does not move when the gripper opens. If it sits on top of the foam, or leans, or sticks up, close on the handle again, lift it just clear of the foam, turn it, and lower it again.

Expected state: the tool is seated in its own cut-out. Its line is ticked. Its slot on the shelf is empty. The pen is in its groove. Both grippers are back and clear.

Step 5: Do the next four tools

Goal: all five tools are seated, and all five lines are ticked.

- Do Step 1 to Step 4 again for line 2. Then for line 3. Then for line 4. Then for line 5.
- The next tool is always the tool on the top line with an empty box. Never skip a line. Never go back up.
- Never lift a tool while another tool is still in a gripper. Never tick a line before its tool is off the shelf.

Check after each tool: the number of ticks on the list, the number of tools in the foam, and the number of empty list slots are all the same. If they are not, stop, find the tool that is out of step, and fix it before the next line.

Expected state: five tools are seated in the foam. Five boxes are ticked. Five list slots are empty. The pen is in its groove. Both grippers are back and clear.

Step 6: End the episode

- Check the list: five ticks, one in each box, none outside a box, and no other marks on the sheet.
- Check the foam: all five cut-outs are filled. Each one holds the tool whose name is beside it. Every tool is seated.
- Check the shelf: the five list slots are empty, all five tags are in place, and the other tools have not moved.
- Check the pen: it lies flat in its groove, tip toward the shelf.

- Check the clipboard: it lies flat on the ledge, list facing up, where it started.
- Check the base and the cart have not moved.
- Fix any failed check before you end.
- Send both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift each tool out of its cut-out by hand. Lay it back flat in its own slot on the shelf, inside the outline, handle to the front, behind its tag.
2. Take the ticked sheet off the clipboard and file it. Put a fresh sheet on the clipboard. It has the same five tools in the same order, and all five boxes are empty.
3. Lay the clipboard flat on the ledge, list facing up, top of the list toward the shelf.
4. Test the pen on scrap paper. Change it if the mark is faint. Lay it in its groove with no cap, tip toward the shelf.
5. Check the foam: all five cut-outs are empty, no edge is torn, and every name beside a cut-out can be read from the front.
6. Check the tags: each tag is behind the right slot and can be read from the front. Change any tag that is worn or loose.
7. Check the other tools on the shelf lie in their own slots and have not been moved.
8. Wipe the shelf and the ledge if they are dusty. Pick up anything on the floor.
9. Check the cart has not moved and both wheel locks are on.
10. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the start timestamp, the violation name, and the SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base or cart moved during the episode

- **Visible cue:** the shelf edge moves in the frame or changes angle, the foam tray shifts in the frame, or the base or the cart rolls, slides, or turns at any time after recording starts.
- **SOP rule broken:** Steps 1 to 6. The base and the cart are parked and locked before recording and stay still for the whole episode.
- **Coaching note:** park them, lock them, push them to test them, then start recording. A base or a cart that moves is a lost episode.

Violation: Leaned on or moved the shelf

- **Visible cue:** a gripper, wrist, or forearm rests on the shelf edge or the shelf above. Or a shelf rocks, or a tag is pushed, or a tool not being pulled slides in its slot.
- **SOP rule broken:** Steps 1 to 6. Nobody leans on, pushes, pulls, or lifts a shelf.
- **Coaching note:** the arm holds itself up. If something moves when you touch it, you touched it too hard.

Violation: Pulled the wrong tool

- **Visible cue:** the tool gripper lifts a tool that is not the one named on the next line. Or it lifts a tool that is not on the list at all.
- **SOP rule broken:** Steps 1.2 and 2.1. Read the next line, find its tag, and pull only that tool.
- **Coaching note:** read first, then reach. The tag tells you the slot. The list tells you the tag.

Violation: Pulled out of order

- **Visible cue:** a tool from a lower line is pulled while a higher line still has an empty box. Or a line is skipped and come back to later.
- **SOP rule broken:** Steps 1.1 and 5. The next tool is always the top line with an empty box. Line 1 to line 5, in order.
- **Coaching note:** top to bottom, one at a time. The list is the order. There is no other order.

Violation: Two tools off the shelf at once

- **Visible cue:** a second tool is lifted while a tool is still in a gripper. Or both grippers hold a tool at the same time.
- **SOP rule broken:** Steps 2 and 5. One tool leaves its slot only after the tool before it is seated in the foam.
- **Coaching note:** one tool from the shelf to the foam before the next one moves. Two at once is twice the ways to drop one.

Violation: Held by the wrong part

- **Visible cue:** the tool gripper closes on the hammer head, the wrench jaws, the pliers jaws, the screwdriver blade or tip, or the tape measure hook instead of the handle. Or it closes on only one handle of the pliers.
- **SOP rule broken:** Step 2.1. Close on the handle in the middle, all the way, before you lift.
- **Coaching note:** handles only. A tool held by its head turns in the gripper and lands wrong in the foam.

Violation: Dragged or knocked on the shelf

- **Visible cue:** the tool is slid along the shelf instead of lifted. Or it hits the tool next to it, the tag, or the shelf edge on the way out. Or it is swung out of the slot.
- **SOP rule broken:** Step 2.2. Lift just off the shelf, then pull straight forward, and touch nothing else.
- **Coaching note:** up a little, then straight out. Nothing on the shelf is touched but the one tool.

Violation: Carried high or over the wrong place

- **Visible cue:** the tool is lifted up toward the shelf above, carried at shelf height, carried over another slot, or carried over the clipboard. Or a wrist or forearm knocks or scrapes the shelf above.
- **SOP rule broken:** Steps 2.2 and 2.3. Lift just off, come straight forward, and carry low and level across to the cart.
- **Coaching note:** out, then across. There is no reason to go high, and the shelf above is right there.

Violation: Tool put down in the middle of a carry

- **Visible cue:** the tool is set down on the ledge, on the shelf edge, on top of the foam, in the wrong cut-out, or on the floor before its line is ticked and its own cut-out is under it.
- **SOP rule broken:** Steps 2.3, 3, and 4. The tool stays in the tool gripper from the shelf until it is seated in its own cut-out.
- **Coaching note:** the tool has one place to go down, and it is the cut-out with its name. Nowhere else is a place.

Violation: Ticked before pulling

- **Visible cue:** the pen marks a box while the tool for that line still lies on the shelf. Or a box below the next line is ticked ahead of its tool.
- **SOP rule broken:** Steps 2 and 3. The tick is made while the tool hangs in the tool gripper over the cart.
- **Coaching note:** pull, then tick. A tick with the tool still on the shelf is a lie on the list.

Violation: Tool laid down without a tick

- **Visible cue:** the tool is lowered into the foam while its line still has an empty box. Or the pen is never lifted for that tool.
- **SOP rule broken:** Steps 3 and 4. Every tool is logged before it is laid down.
- **Coaching note:** tick, then foam. The clipboard is the record. A tool in the foam with no tick was never checked out.

Violation: Tick on the wrong line

- **Visible cue:** the pen marks a box on a line other than the next line. Or the mark lands on a line that is already ticked.
- **SOP rule broken:** Step 3.3. The tick goes in the box on the next line and no other line.
- **Coaching note:** find the top empty box before the pen comes down. The pen goes where the eye already is.

Violation: Bad tick

- **Visible cue:** the mark lands outside the box, or runs over the box edge onto the name. Or there is more than one mark in a box. Or the mark is a scribble or a cross-out. Or the mark is too faint to read.
- **SOP rule broken:** Step 3.3. One short mark inside the box. Press lightly. Do not scribble. Do not cross out.
- **Coaching note:** one short mark, inside the lines, dark enough to see. The list must read clean from the front.

Violation: Pen not put back right

- **Visible cue:** the pen is laid on the clipboard, on the foam, on the shelf, or left in the gripper. Or it is dropped. Or it is put back tip away from the shelf. Or the clipboard is pushed or lifted while the pen is picked up or put back.
- **SOP rule broken:** Steps 3.2 and 3.4. Lift the pen from its groove, tick, and lay it back flat in its groove, tip toward the shelf. Do not touch the clipboard.
- **Coaching note:** the pen lives in the groove. It comes out for one tick and goes straight back.

Violation: Laid in the wrong cut-out

- **Visible cue:** the tool is lowered into a cut-out whose name is not the tool's name. Or it is pushed into a cut-out whose shape does not match.
- **SOP rule broken:** Step 4.1. Read the name beside the cut-out and check the shape before you lower.
- **Coaching note:** name, shape, then down. A hole that does not fit is not the hole.

Violation: Tool not seated

- **Visible cue:** the tool ends up on top of the foam, leaning in its cut-out, upside down, handle at the wrong end, or with a part sticking up above the foam. Or it moves when the gripper opens.
- **SOP rule broken:** Steps 4.2 and 4.3. Line it up over the shape, lower it straight down until it rests on the bottom, and only then let go.
- **Coaching note:** straight down, then watch it settle before you open. If it rocks, it is not in.

Violation: Forced into the foam

- **Visible cue:** the tool is pushed down hard, hammered in, or wiggled hard against the foam. Or the foam tears, dents, or lifts at an edge.
- **SOP rule broken:** Step 4.3. Lower it in. Do not push it. If it will not go in, lift it, turn it a little, and lower it again.
- **Coaching note:** the foam is soft and the fit is close. Fix the angle, not the force.

Violation: Grippers swapped jobs

- **Visible cue:** the gripper holding a tool touches the pen. Or the pen gripper touches a tool or the clipboard. Or a tool is passed from one gripper to the other. Or the gripper carrying a tool changes in the middle of a tool.
- **SOP rule broken:** Gripper rules and Steps 2 to 4. One gripper carries the tool the whole way. The other gripper works the pen and nothing else. Nothing is passed.
- **Coaching note:** one hand carries, the other hand ticks. Nothing swaps, not even once.

Violation: Touched a tool not on the list

- **Visible cue:** a gripper closes on, lifts, pushes, or slides a tool that lies in one of the other slots.
- **SOP rule broken:** Materials checklist line 5 and Steps 1 to 6. Only list tools are touched.
- **Coaching note:** if its name is not on the list, it is not part of the task. Leave it where it lies.

Violation: Something dropped or knocked over

- **Visible cue:** a tool or the pen slips out of a gripper and falls on the shelf, the cart, or the floor. Or the clipboard, a tag, or another tool is knocked by an arm going past.
- **SOP rule broken:** Steps 1 to 6. One thing moves at a time, and each one is put down under control.
- **Coaching note:** close all the way before you lift, and put it down before you let go.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails, and the episode carries on with no retry. For

example, the tool leans in the gripper, the tick misses the box, the tool sits on top of the foam, or the counts do not match.

- **SOP rule broken:** Steps 1 to 5. Do the retry written under each failed check before you move on.
- **Coaching note:** a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

- **Visible cue:** the episode ends with a tool still on the shelf, a box empty, a tool not seated, a tool in the wrong cut-out, the pen out of its groove, or the clipboard moved. Or an arm is not home, or a gripper is not fully open.
- **SOP rule broken:** Step 6. Check the list, the foam, the shelf, the pen, the clipboard, the base, and the cart. Then send both arms home with the grippers open and stop recording.
- **Coaching note:** check first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Camera cannot see the tick boxes, the name tags, or the cut-outs, so nobody can tell if a line was ticked, which slot a tool came from, or if a tool is seated.
- Hardware fault on an arm: gripper failure, drift, a crash caused by the controller, or a motor error.
- Base fault: a lock that will not hold, or a base that rolls, slides, or turns with the lock on.
- Cart fault: a wheel lock that will not hold, or a cart that rolls with both locks on.
- Crib fault: a list tool missing from its slot at the start, a list tool in the wrong slot at the start, or a shelf that rocks under a correct pull.
- Tag fault: a tag that is missing, unreadable, loose, or names a different tool from the one in the slot.
- List fault: a sheet with fewer or more than five lines, a name that matches no tag, a name printed wrong, a box already ticked at the start, or a sheet that slides on the clipboard.
- Pen fault: a pen that is dry, makes a faint mark under a light touch, or rolls out of its groove on its own.
- Foam fault: a cut-out that does not fit its tool under a correct set down, a torn or lifted edge, or a name beside a cut-out that is worn off or wrong.
- Tool fault: a handle that comes loose in a correct grip, a tool that comes apart when lifted, or a tape measure whose blade runs out on its own.

- A place turns out to be too far for its arm with the base and the cart parked correctly, so a list slot, the pen, a tick box, or a cut-out cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Read the next line on the list
2. Find the tool's slot on the shelf
3. Lift the tool out of its slot
4. Carry the tool to the cart
5. Pick up the pen
6. Tick the box on the list
7. Put the pen back in its groove
8. Lay the tool in its cut-out
9. Send both arms home and end the episode

Subtasks 1 to 8 happen once for each of the five tools, in list order. Subtask 9 happens once, at the end.